LECTURE ACTIVITY 17: GOING ONCE, GOING TWICE... – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Learning Objectives:

By the end of this activity, you should be able to

1) Distinguish between independent samples and paired samples in terms of the study design.
2) State the null and the alternative hypotheses in terms of comparing population means, in a paired design, for two categories of the explanatory variable (i.e., μ_d).
3) Implement the 3S strategy: find a statistic, simulate, and compute the strength of evidence against observed study results happening by chance alone.
4) Conduct a simulation-based analysis when evaluating claims about paired data for a quantitative response.
5) Find, interpret, and/or evaluate the standardized statistic and the p-value for a test of two population means in a paired design.
6) Conduct a theory-based analysis when evaluating claims about two population means in a paired design.
7) Identify whether the validity conditions for a paired *t*-test are met.
8) Use the theory-based method to find and interpret a 95% confidence interval for the population mean of the differences.
9) State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the standardized statistic and/or p-value and the study design.

Step 1: Formulate a Research Question

Two popular formats of online auctions are:

- **Dutch Auctions**: the item for sale starts at a very high price, and then price is gradually lowered until someone finds the price low enough to buy
- **First-Price Sealed Bid Auctions**: each bidder submits a single sealed bid before a particular deadline; then, after the deadline, the person with the highest bid wins

Does the format of auction used (Dutch vs. First-Price Sealed Bid) affect the final sale price? Dr. David Lucking-Riley, formerly at Vanderbilt University, devised a study[1] to compare different types of online auctions. Here is a slightly simplified version of the experimental setup he used: he placed pairs of identical *Magic: The Gathering* cards up for auction, one in a Dutch auction and the other in a first-price sealed bid auction. He then looked at the difference in the prices (in USD) he received on the pair, and repeated this for a total of 88 pairs.

[1] Reiley, David H. "Field Experiments on the Effects of Reserve Prices in Auctions: More Magic on the Internet." *The RAND Journal of Economics*, vol. 37, no. 1, 2006, pp. 195–211. JSTOR, http://www.jstor.org/stable/25046235.

Step 2: Design a Study and Collect Data

Before we look at the data that was collected and start the analysis, let's make sure you understand the study design.

Question 1

(a) Explain why the price data should be analyzed using **paired** samples as opposed to two independent samples.

Solution: The prices should be analyzed as paired samples because each Dutch-auction price is matched with the first-price sealed-bid price for an identical trading card. The two prices within each pair are connected, rather than coming from unrelated cards.

(b) Identify the two variables in this study. Then, classify each of them as either categorical or quantitative, and identify which variable is the explanatory variable and which is the response variable.

- **Variable 1:** **Auction Format (Dutch or first-price sealed bid)**
  - Classification: √ **Categorical** ◯ Quantitative
  - Type: √ **Explanatory** ◯ Response

- **Variable 2:** **Selling Price (in USD)**
  - Classification: ◯ Categorical √ **Quantitative**
  - Type: ◯ Explanatory √ **Response**

(c) State the null and alternative hypotheses in symbols and math only.

Solution: $H_0 : \mu_d = 0$ vs. $H_A : \mu_d \neq 0$

(d) Define the symbol(s) you used in your hypotheses above.

Solution: The symbol μ_d represents the population mean difference in price, where each difference is calculated as Dutch-auction price minus first-price sealed-bid price for a pair of identical cards.

(e) Is this an observational study or an experiment?

◯ Observational Study
√ **Experiment**

Solution: Dr. Lucking-Riley controlled which (of the two identical cards) went to which auction.

Step 3: Explore the Data

We'll begin (as always) by exploring the data bit more; in particular, we'll use the **Matched Pairs applet**.

1) Navigate to the Matched Pairs applet

2) In the dropdown menu, select "**Auction**"

3) Press "**Use Data**"

Notice that the applet graphs the prices in each group, along with the means and standard deviations for the two groups. The red dots represent the prices for the Dutch auction and the blue dots represent the prices for the First Price auction.

Question 2

(a) The sample mean price for the Dutch auction is: **$5.613**

(b) The sample mean price for the first-price sealed bid auction is: **$4.779**

The applet also provides a dotplot with black dots representing the differences in the prices between Dutch and first-price auction in the sample. Please note that these difference values can be negative numbers because you are looking at change or difference in prices. As seen in Chapter 6, we can summarize the quantitative data (price paid) using means. But, because the data are paired, we will instead look at the **mean of differences** in price between the two types of auction (Dutch or first-price sealed bid).

Question 3

What are the mean, standard deviation, and sample size of the differences in the two prices?

$n =$ **88** $\bar{x}_d =$ **$0.384** $s_d =$ **$0.675**

Step 4: Draw Inferences Beyond the Data

We once again arrive at our perennial question: is the sample mean of differences in prices that we observed something that could have plausibly happened by chance, if, on average, there is no difference between the Dutch and First-price sealed big auctions? Enter - the **3S Method!**

Statistic

One and done - this is our value from Question (3) above!

Simulation

Once again, I think it will be useful to run through the *logic* of the simulation before actually carrying it out. Our null hypothesis is that the auction type has no effect on the final sale price. If this is true, then

it actually didn't matter which of the two cards in any given pair got assigned to the Dutch auction and which got assigned to the first-price sealed bid auction - the two cards would have sold for the exact same prices if their auction type had been swapped. So, our simulation should then run through the "alternate universes" where the *assignments* of each card within each pair was (potentially) swapped.

This is precisely what our simulation will do - it will flip a coin to determine whether or not to swap the values in a pair. If the coin lands "heads," swap the prices - otherwise, don't swap.

Question 4

Why should we *not* use the index card simulation we used last week (in Lecture Activities 15 and 16, "Lingering Effects of Sleep Deprivation" and "Plate Up!", respectively)?

Solution: Conducting our simulation like this would break the pairing - we want to ensure that each pair of prices (paired because they are from identical cards) remains "linked" throughout our simulation.

We'll use the Matched Pairs applet to conduct our simulation.

1) Check the box next to **Randomize**.

2) Click on the **Randomize** button.

- Notice that the applet displays the re-randomized data according to the result of the coin toss (the colors show you the original column for each observation, so you should see a mix in each group).
- The could-have-been value for the mean difference is added to the Average Difference graph.

3) Update the number of times to **Randomize** to 9,999 (for a total of 10,000 repetitions), uncheck **Animate**, and press **Randomize**.

Question 5

What does each dot represent in the simulated null distribution?

Solution: Each dot in the simulated null distribution represents the sample mean of the differences in prices.

Question 6

(a) Use the applet to find and report the simulation-based *p*-value. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) *p*-value.

- My simulation-based *p*-value is: **0.000** **(answers may vary)**

Here is my Screenshot:

Solution:

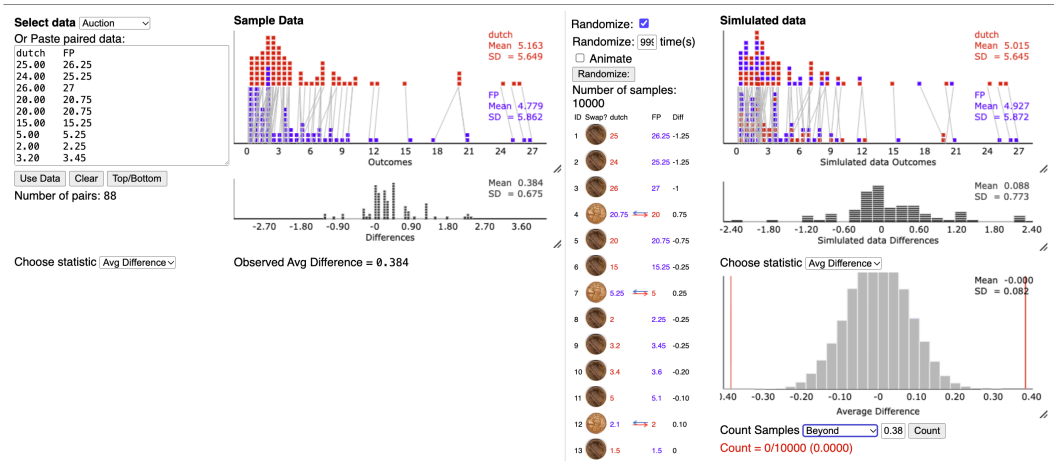


Figure 1: Screenshot from the Matched Pairs applet

(b) ***Evaluate*** this *p*-value: How much evidence does the sample data provide against the null hypothesis? Choose one

- ◯ Not much evidence against the null hypothesis
- ◯ Moderate evidence against the null hypothesis
- ◯ Strong evidence against the null hypothesis
- √ **Very strong evidence against the null hypothesis**

Theory-Based Approach

Notice that the simulated null distribution is bell-shaped. This is no coincidence! A theory-based method exists that predicts this to occur when certain validity conditions are met.

Concept Overview

Validity Conditions for a Theory-Based Approach: Theory-based methods of inference will work well for paired data if the distribution of differences has a symmetric distribution, or you have at least 20 pairs (i.e., at least 20 differences) and the distribution of the sample differences is not strongly skewed. This test is known as a **paired *t*-test**.

Question 7

Explain why the validity conditions are met in this study.

Solution: Yes. The validity conditions are met because there are 88 paired differences, which is much more than 20, and the sample distribution of the differences in prices is not strongly skewed. Therefore, a theory-based approach is valid.

To conduct our paired *t*-test, we'll use.... an ***old* applet**? Yup - the Theory-Based Inference applet!

1) Navigate to the Theory-Based Inference applet, and choose "**One Mean**" from the dropdown menu.

2) Enter the sample size, sample mean, and sample standard deviation for the differences in prices between the two auctions (see Question 3 above).

3) Press "**Calculate**"

4) Check the box for "**Test of Significance**".

5) Enter the appropriate information for the null and alternative hypotheses. Make sure the appropriate sign for the alternative hypotheses is chosen.

6) Press "**Calculate**."

Question 8

(a) Report the theory-based p-value and the standardized statistic. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) p-value.

- My theory-based *p*-value is: **0.000**
- My theory-based standardized statistic is: **5.34**

Here is my Screenshot:

Solution:

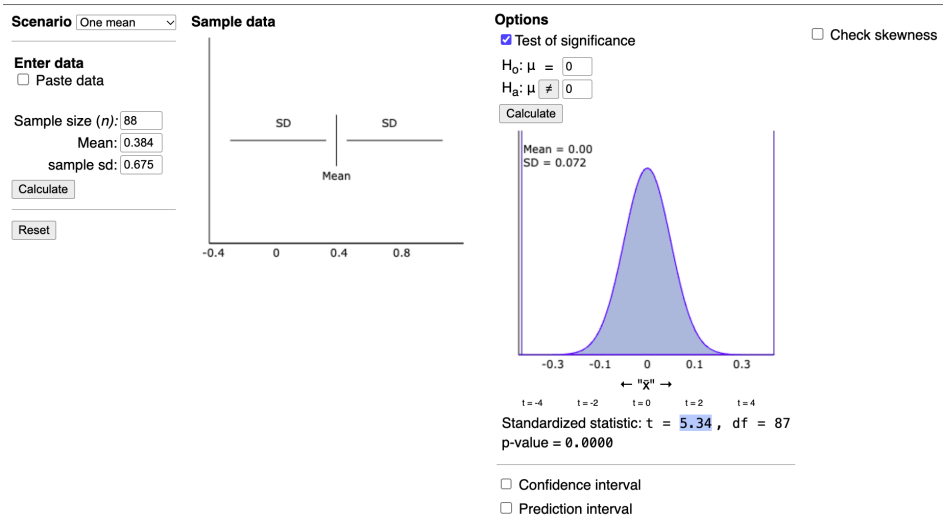


Figure 2: Screenshot from the Theory-Based Inference applet

(b) Is the theory-based *p*-value similar to the simulation-based *p*-value? Is this to be expected?

Solution: They are similar, which is to be expected since the validity conditions are met.

Step 5: Formulate Conclusions

It's time to start formulating our conclusions!

Question 9

Significance. Based on your analysis, give an overall conclusion in context. Make sure you refer to the parameter of interest in your conclusion

Solution: We have very strong evidence that the population mean of the differences in selling price between Dutch and first-price bid auction is not 0. In other words, we have sufficient evidence of is a difference in population mean prices paid using the two different auction formats, in the population.

Question 10

Generalization. To which population can you generalize the results of this study? Other kinds of cards? Other types of items? Anything sold in an auction format on the Internet? How are you deciding?

Solution: The results should be generalized cautiously to Magic: The Gathering cards and online auction situations similar to those used in this study. The cards were not described as a random sample of all trading cards or all items sold online, so the results should not automatically be generalized to other types of cards, other products, or every Internet auction. It might be that certain type of cards are available more often in one type of auction than the other.

Question 11

Causation. Can you make a causal conclusion? If yes, what causes what? If not, how are you deciding?

Solution: A cause-and-effect conclusion is reasonable for the cards and auction conditions studied because the researcher imposed the auction formats on pairs of identical cards. Pairing identical cards helps control for differences in the items being sold. Therefore, the auction format can be viewed as causing the difference in population mean selling price under similar conditions.

Estimation

Recall that the Theory-Based Inference applet can also produce confidence intervals for the parameter of interest.

Question 12

Before using the applet: do you expect the 95% confidence interval to contain zero?

Solution: No. Because the hypothesis test produced a very small p-value and rejected the null value of 0, the corresponding 95% confidence interval should not contain 0.

Question 13

(a) Use the applet to calculate and report the 95% confidence interval.

Solution: ($0.2410, $0.5270). Remember to include units.

(b) Interpret the confidence interval.

Solution: We are 95% confident that, for auctions similar to those in the study, the population mean selling price for a Dutch auction is between $0.241 and $0.527 higher than the population mean selling price for Magic: The Gathering cards similar to those used in this study. Remember: state which (if either) is higher!

(c) Does the 95% confidence interval agree with your answer to Question 12 above? Explain.

Solution: Yes, because the entire interval, ($0.241, $0.527), is above 0. Therefore, a population mean of the differences in prices of 0 is not a plausible value, which agrees with the small p-value found earlier and the rejection of the null hypothesis.

Step 6: Looking Back and Ahead

Question 14

Critique the study.

Solution: A major strength is the paired design: each Dutch-auction card was matched with an identical card in the first-price sealed-bid auction, which reduces variability caused by differences among cards. The sample also included 88 pairs, which is large enough for the theory-based method. Limitations include the use of only *Magic: The Gathering cards*, possible differences in bidders or auction timing, and the lack of a random sample of all cards or online auctions. Replication with other products, bidders, and auction settings would improve generalizability.